Store Intelligence Challenge — Evaluation Framework

1. Context and Evaluation Philosophy

This challenge is intentionally designed as an end-to-end system problem — not a model-building exercise.
Candidates are expected to start from raw CCTV footage and build a complete pipeline that produces meaningful business metrics such as store conversion rate. As outlined in the challenge briefly, the goal is not perfect detection accuracy, but the ability to:

- Decompose a real-world ambiguous problem
- Build a working system with reasonable assumptions
- Handle known edge cases (re-entry, staff movement, occlusion, etc.)
- Translate raw signals into business-relevant insights

Accordingly, this evaluation framework prioritizes:

- Functional correctness over theoretical completeness
- Engineering judgment over model complexity
- Clarity of reasoning over volume of implementation

A candidate is not expected to build a perfect system. A strong candidate is one who builds a system that works, makes reasonable trade-offs, and can clearly explain those decisions.

2. Evaluation Process Overview

The evaluation is conducted in three stages:

Stage	Purpose
Acceptance Gate	Eliminate incomplete or non-functional submissions
Structured Scoring	Evaluate technical quality and system correctness
Final Shortlisting	Identify top 30 candidates based on score and consistency

Each submission is evaluated independently by two reviewers.

3. Acceptance Gate (Mandatory)

Submissions must satisfy the following:

Check	Requirement
System Execution	docker compose up runs without manual intervention
API Availability	/Metrics endpoint returns a valid response
Event Generation	Detection pipeline produces structured events
Documentation	DESIGN.md and CHOICES.md are present and non-trivial

Stability	System does not crash during basic execution

Failure in any of the above results in rejection prior to scoring.

4. Reviewer Evaluation Approach

Each submission is evaluated within a structured time window to ensure consistency:

Time	Activity
2 minutes	Run system and verify API
2 minutes	Inspect generated events
3 minutes	Validate API outputs
2 minutes	Review DESIGN.md and CHOICES.md
1 minute	Assign scores

05 Scoring Framework (100 Marks)

5.1 Detection Pipeline (30 Marks)

Criteria	Strong	Moderate	Weak
Entry/Exit:	Close to actual counts (within reasonable error margin)	Noticeable deviation	Unreliable
Accuracy:	Handles re-entry, staff, and group entry correctly	Partial handling	Not handled
Edge Case Handling:	Structured, complete, and consistent events	Minor inconsistencies	Poor or incomplete

VALIDATION APPROACH
- Run a sample clip and compare approximate entry counts with system output
- Inspect event schema for completeness and consistency

5.2 API and Business Logic (35 Marks)

Criteria	Strong	Moderate	Weak
Endpoint/Correctness:	All endpoints return correct and consistent results	Minor inconsistencies	Incorrect outputs
Funnel Logic:	Session-based, no double counting	Basic	Missing
Anomaly Detection:	Logical and meaningful	Basic	Missing

VALIDATION APPROACH
/metrics returns logically consistent values
/funnel shows expected drop-off behavior

5.3 Production Readiness (20 Marks)

Criteria	Strong	Moderate	Weak
Deployment:	Runs seamlessly with minimal setup	Requires minor effort	Not functional
Observability:	Comprehensive (logs, metrics, tracing)	Partial	Missing
Testing:	Covers key scenarios and edge cases	Limited coverage	No testing

5.4 Engineering Thinking and Decision Making (15 Marks)

Criteria	Strong	Moderate	Weak
CHOICES.md:	Clear trade-offs and justification	Some reasoning	Generic
DESIGN.md:	Clear system architecture	Basic explanation	Unclear
Reasoning Depth:	Demonstrates independent thinking	Limited depth	Superficia l

Reviewers should assess whether the candidate demonstrates ownership of decisions or relies on generic explanations.

06 Integrity Check

To ensure authenticity of submissions:

Check	Action
Hardcoded outputs suspected	Score capped at 50
Outputs do not vary with input	Score capped at 50
Lack of real computation	Score capped at 50

07 Final Scoring and Shortlisting

Final score is computed as:

Total = Detection (30) + API (35) + Production (20) + Thinking (15)

Shortlisting guidelines:

Score Range	Interpretation
85+	Strong candidate
70-85	Suitable for interview
60-70	Above Average

Top 30 candidates are selected based on score and consistency across reviewers.

8. Tie-Breaking Criteria

In cases of similar scores, preference is given to candidates who demonstrate:
Better handling of edge cases
Cleaner and more maintainable system design
Stronger understanding of the underlying business metric Clear and structured reasoning

9. Evaluation Principle

This framework is designed to identify candidates who can:

- Build working systems under constraints
- Handle real-world ambiguity
- Make and justify engineering decision